

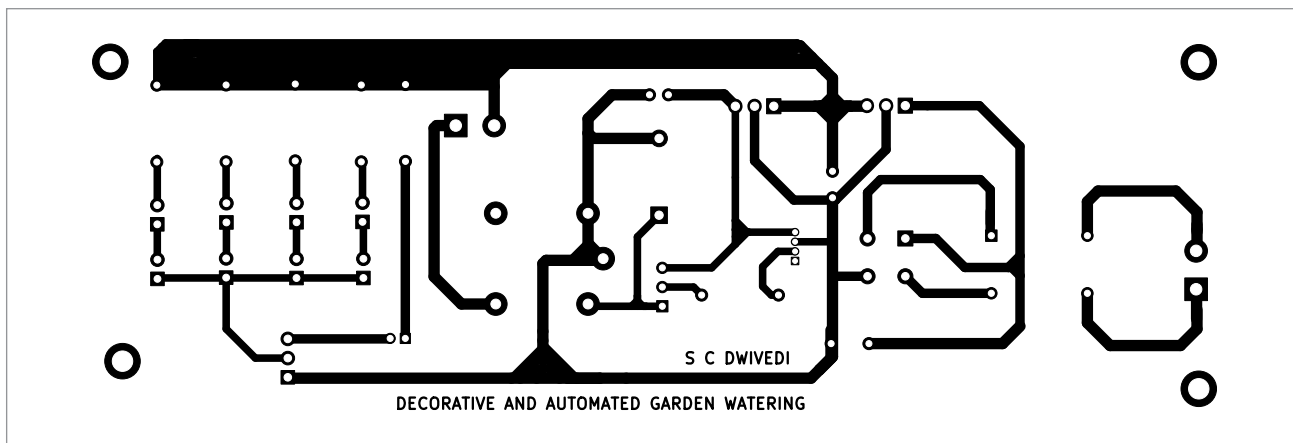


Fig. 3: Actual-size PCB layout

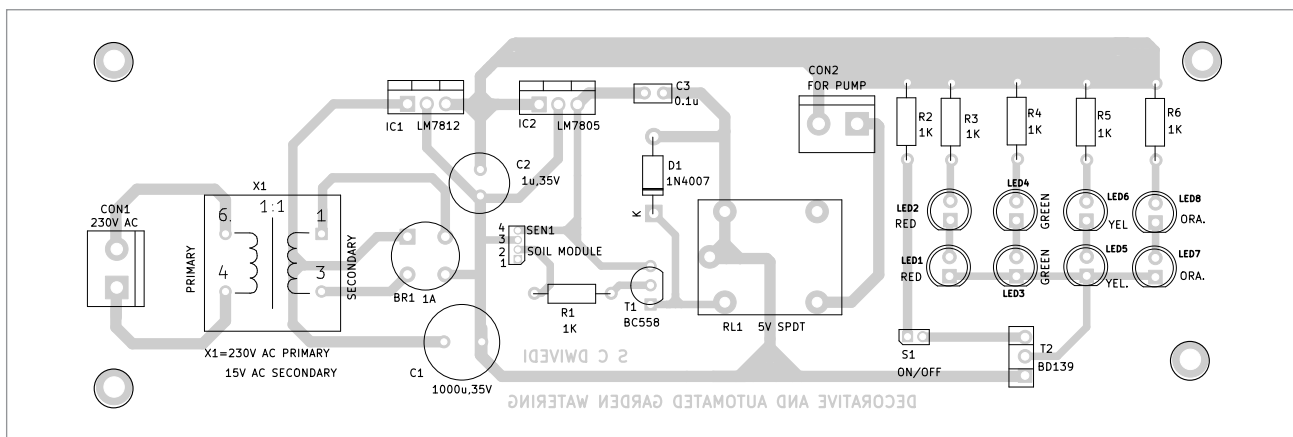


Fig. 4: Components layout

output from X1 is fed to the bridge rectifier (BR1). Capacitor C1 filters the voltage ripples, and this filtered output is supplied to the voltage regulator LM7812. IC1 provides a 12V regulated DC output, which is further connected to the input of IC2, the 12V water pump, and the eight LEDs. IC2 generates a 5V regulated DC output to drive the soil sensor and relay RL1. Capacitors C2 and C3 filter any remaining voltage ripples.

The circuit is divided into two parts: automated water gardening and the decorative lighting system.

Automated water gardening

The soil sensor module and its probe form the heart of this device.

The module has two pins for the probe and four pins for output and power supply. Of the four pins, two serve as VCC and GND, while the remaining two are for digital output (D0) and analogue output (A0). The analogue output is not utilised in this application.

The probe's resistance is inversely proportional to the soil moisture content: higher moisture results in better conductivity and lower resistance, while lower moisture results in poor conductivity and higher resistance. When water is added to the soil, the digital output pin changes from low (0V) to high (5V) due to the onboard comparator op-amp. The sensitivity of this digital output can be

PARTS LIST

Semiconductors:

- IC1 - LM7812, 12V voltage regulator
- IC2 - LM7805, 5V voltage regulator
- T1 - BC558 PNP transistor
- T2 - BD139 NPN transistor
- BR1 - 1A bridge rectifier
- D1 - 1N4007 rectifier diode
- LED1-LED8 - 5mm LEDs of different colour

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R6 - 1-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 1 μ F, 35V electrolytic
- C3 - 0.1 μ F, ceramic disk

Miscellaneous:

- S1 - On/off switch
- CON1-CON2 - 2-pin connectors
- RL1 - 5V SPDT relay
- X1 - 230V AC primary to 15V, 1A secondary transformer
- SEN1 - Soil sensor module with probe and jumper wires